Ethanol Blending in Gasoline – Spain

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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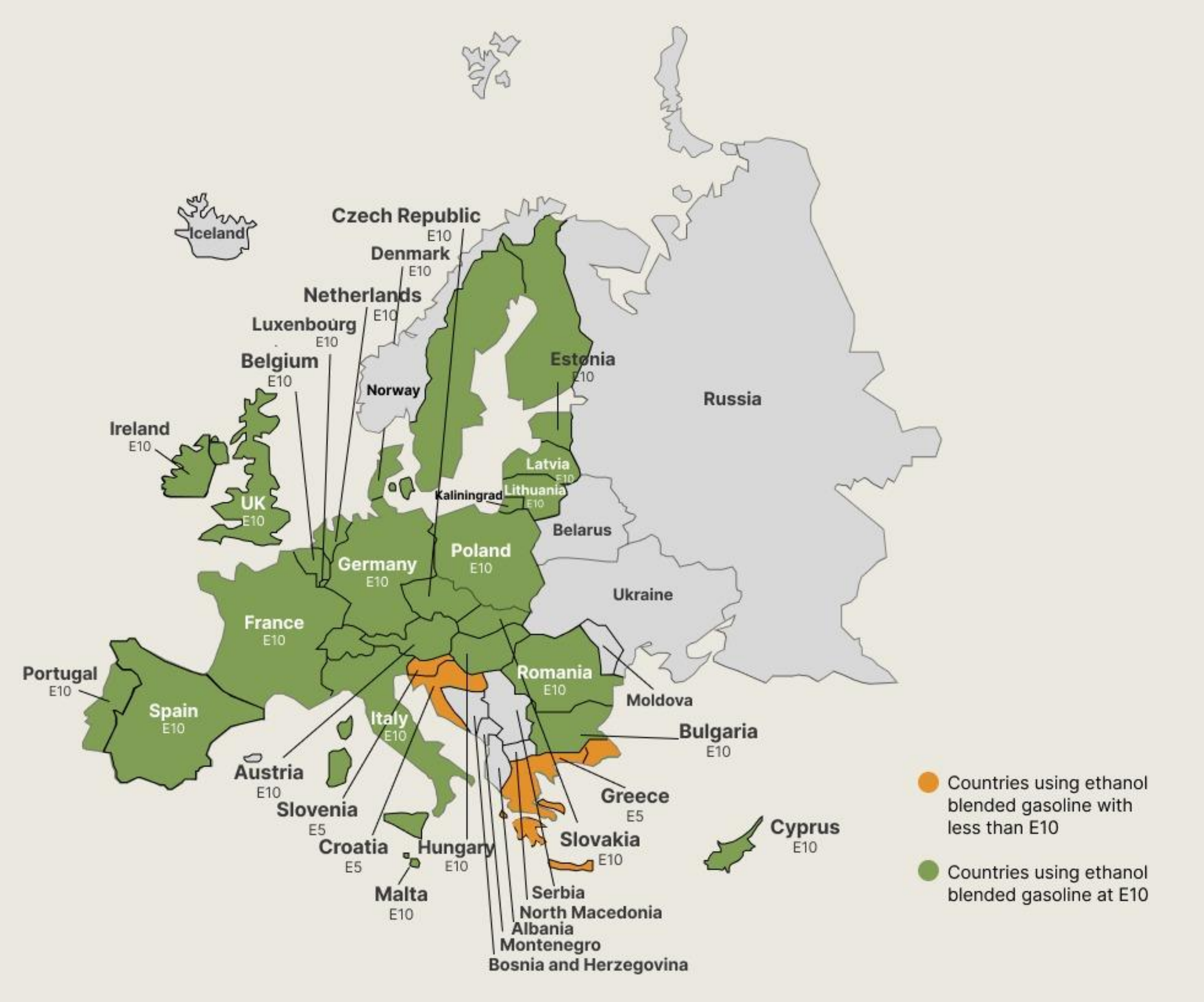
Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

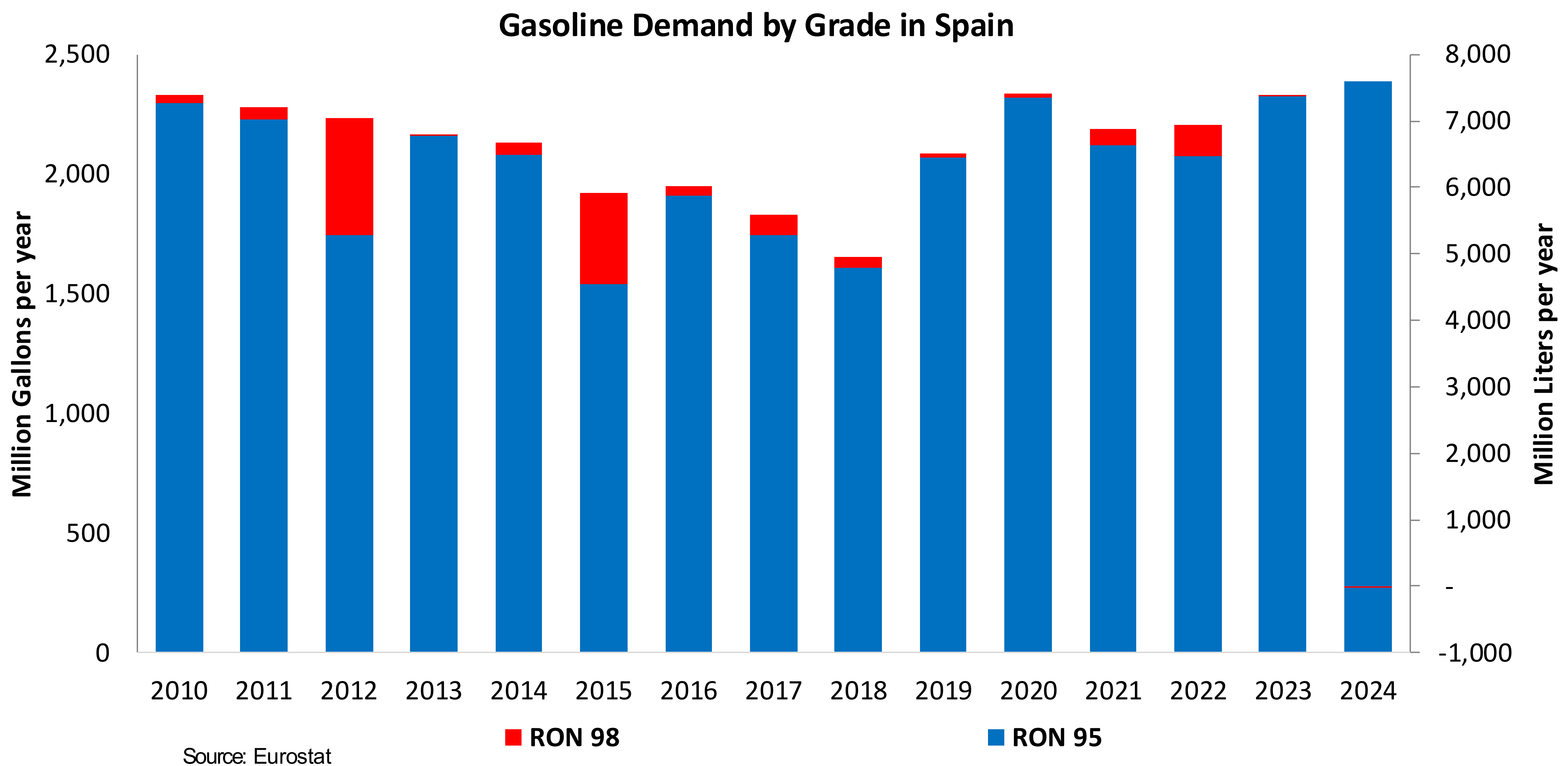
Ethanol Gasoline Blending in Spain

In 2024, gasoline consumption in Spain was 7,577 million liters (2,002 million gallons) and growth rate was 3.1%. Gasoline production and net imports were 11,890 and -3,071 million liters (3,141 and -811 million gallons) correspondingly. Spain complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.0 RON.

In 2024, ethanol consumption in Spain was 253 million liters (67 million gallons) representing 3.3% of gasoline demand. Ethanol production and Net Imports were 545 and -292 million liters (67 and -77 million gallons) correspondingly. Ethanol sources are Corn 60%, Cereals 25% and Sugar Beet/Wine 40%.

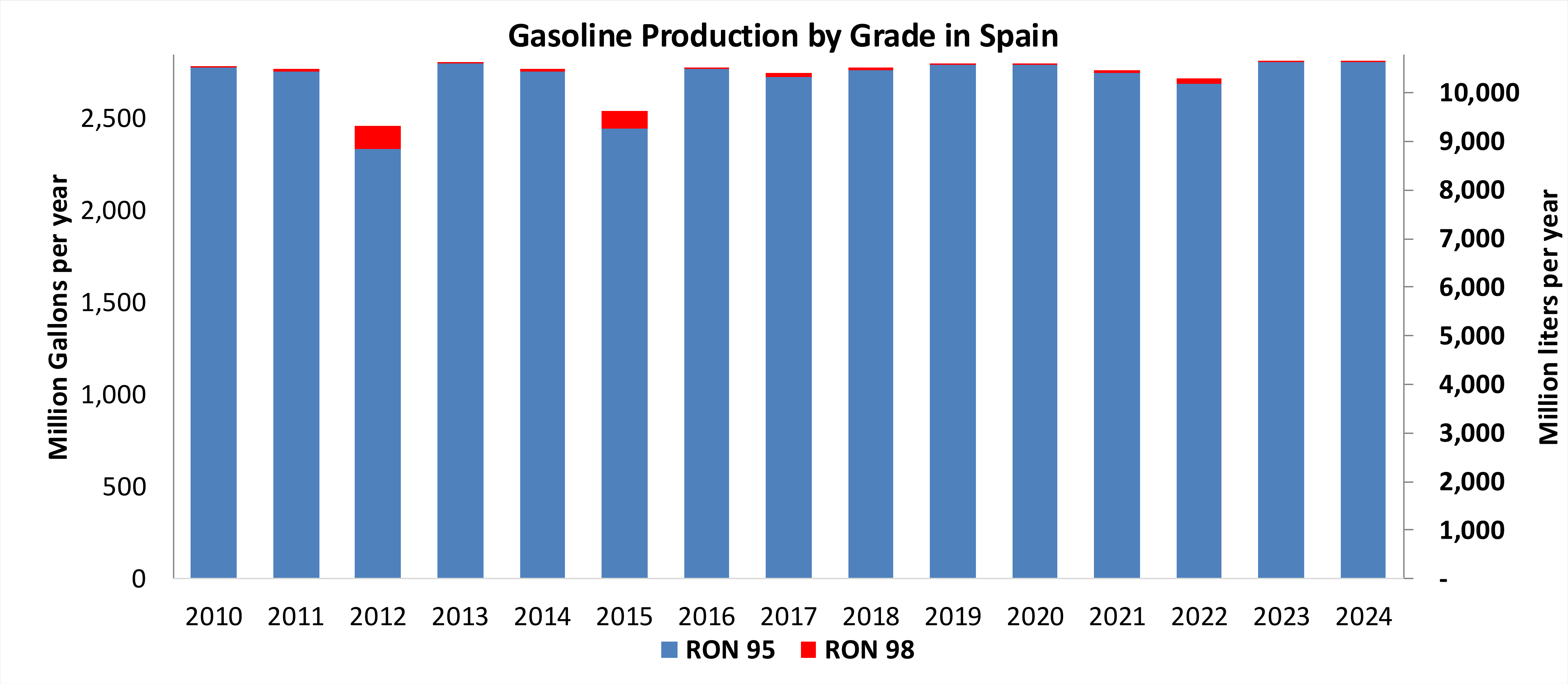
Source: Eurostat, European Commision

Gasoline Demand in Spain



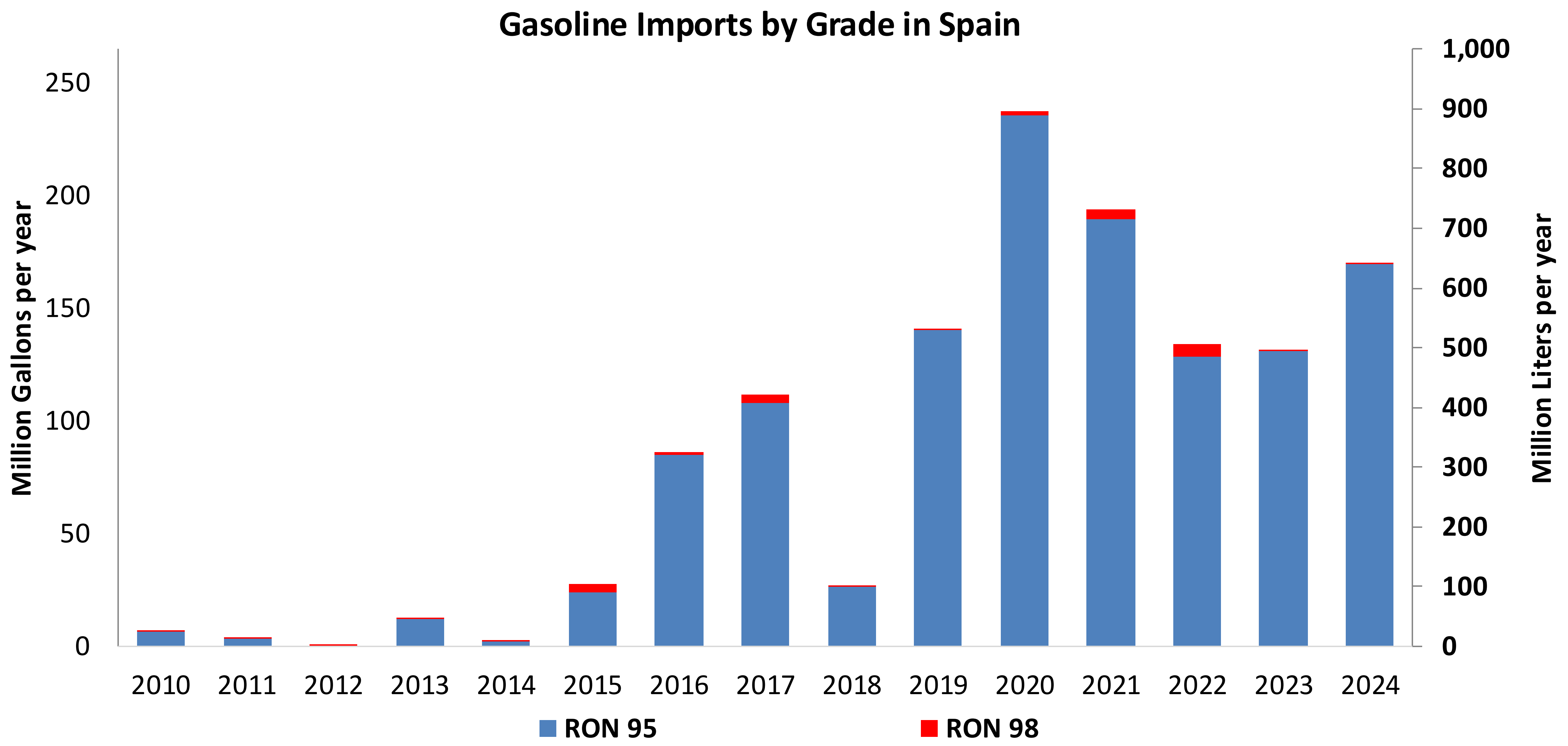
Source: Eurostat

Gasoline Production in Spain



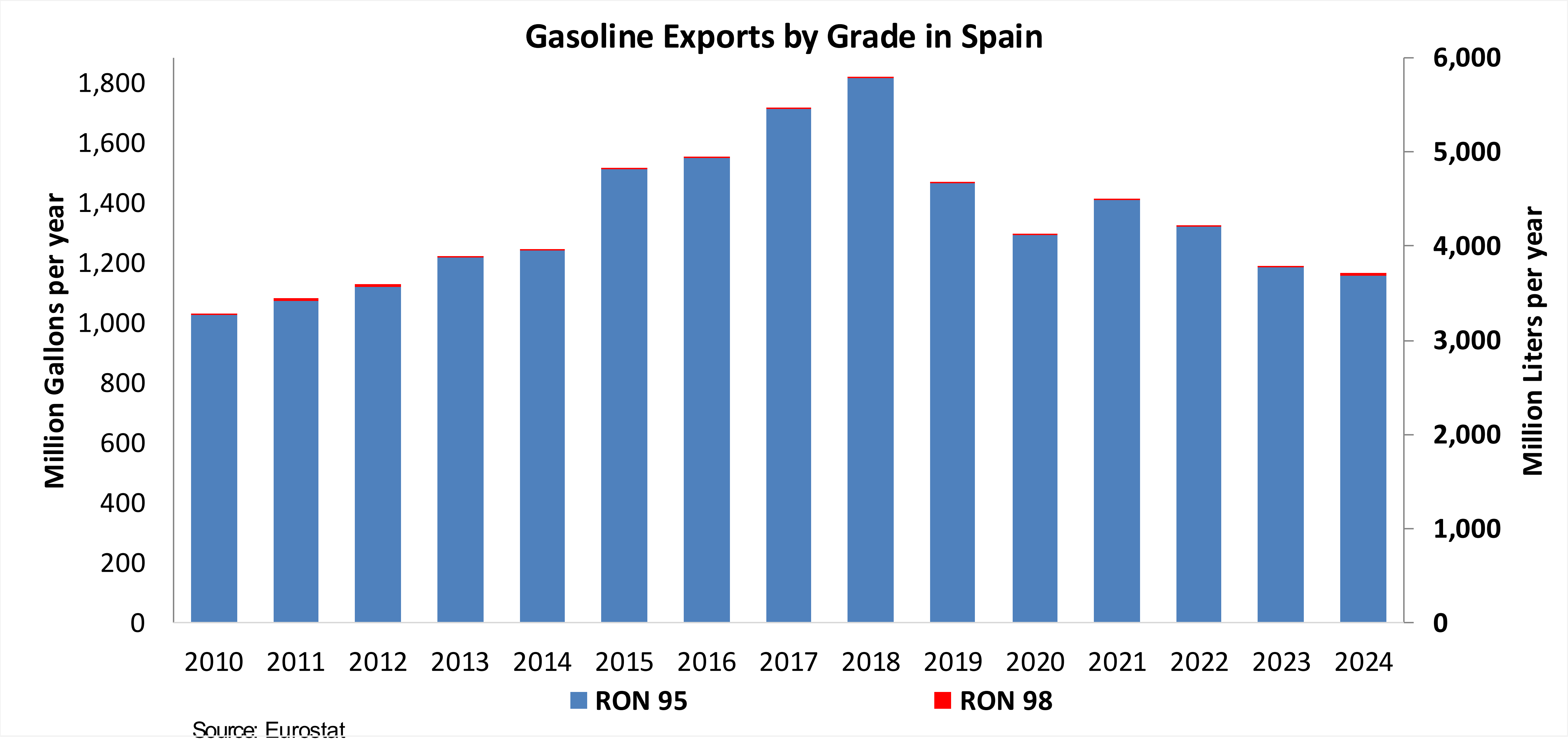
Source: Eurostat

Gasoline Imports in Spain

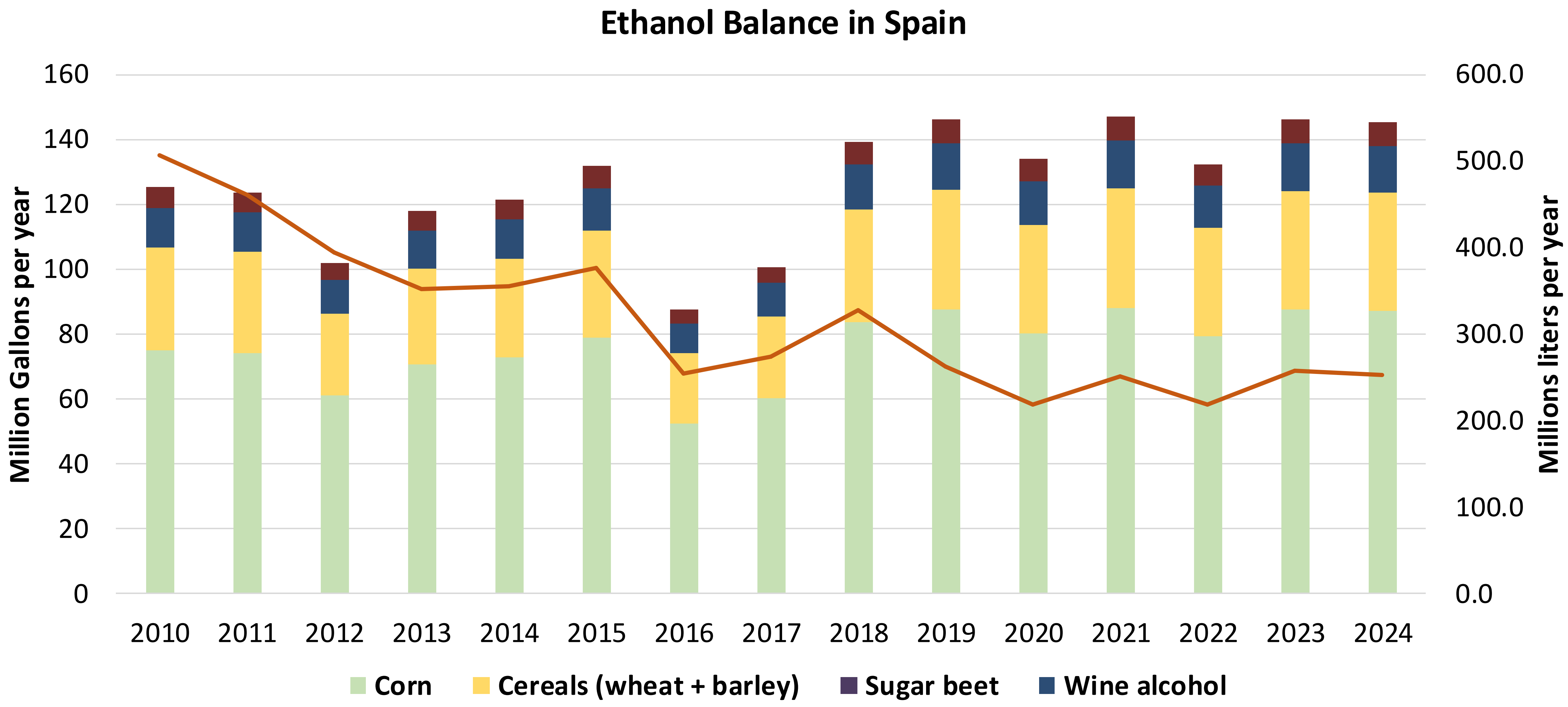


Source: Eurostat

Gasoline Exports in Spain



Ethanol Balance in Spain



Source: Eurostat

Gasoline Quality in Spain

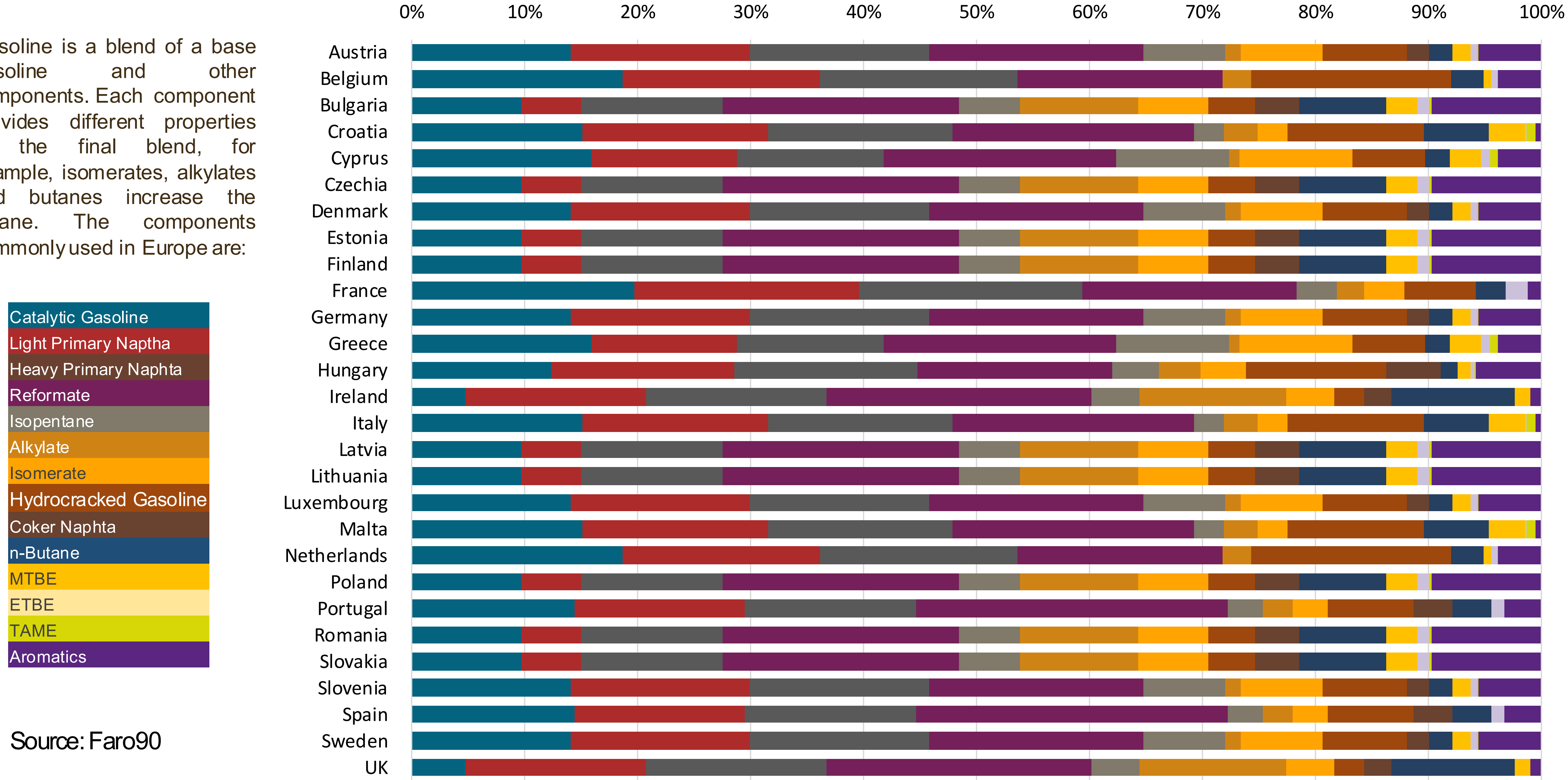
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.5							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

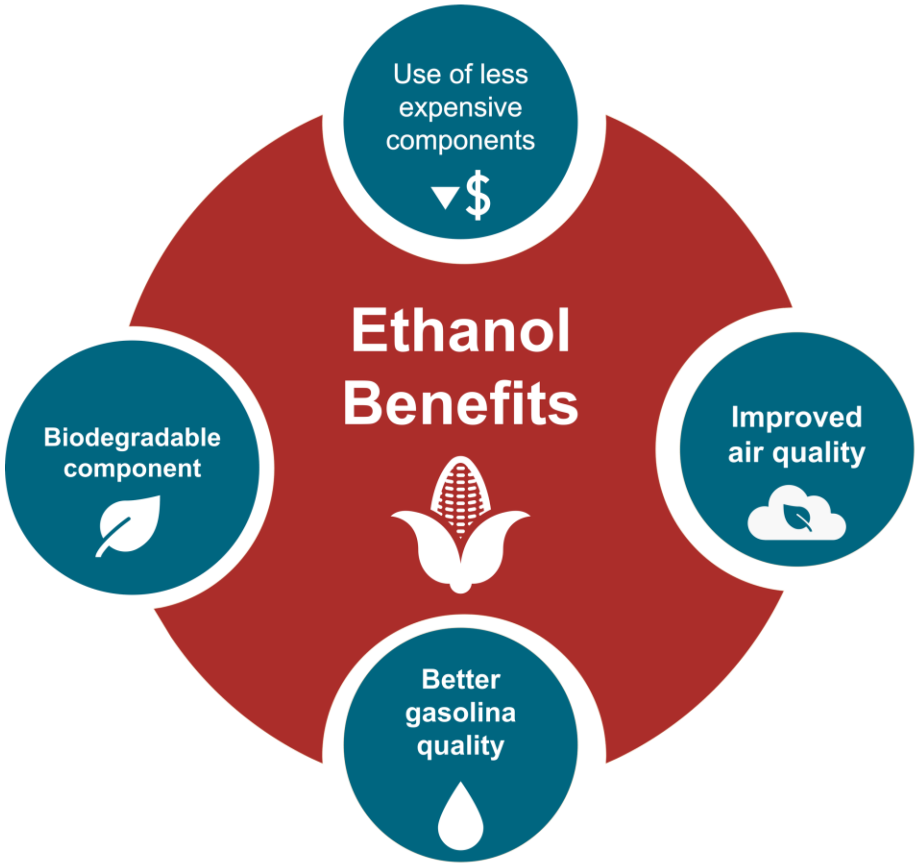
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

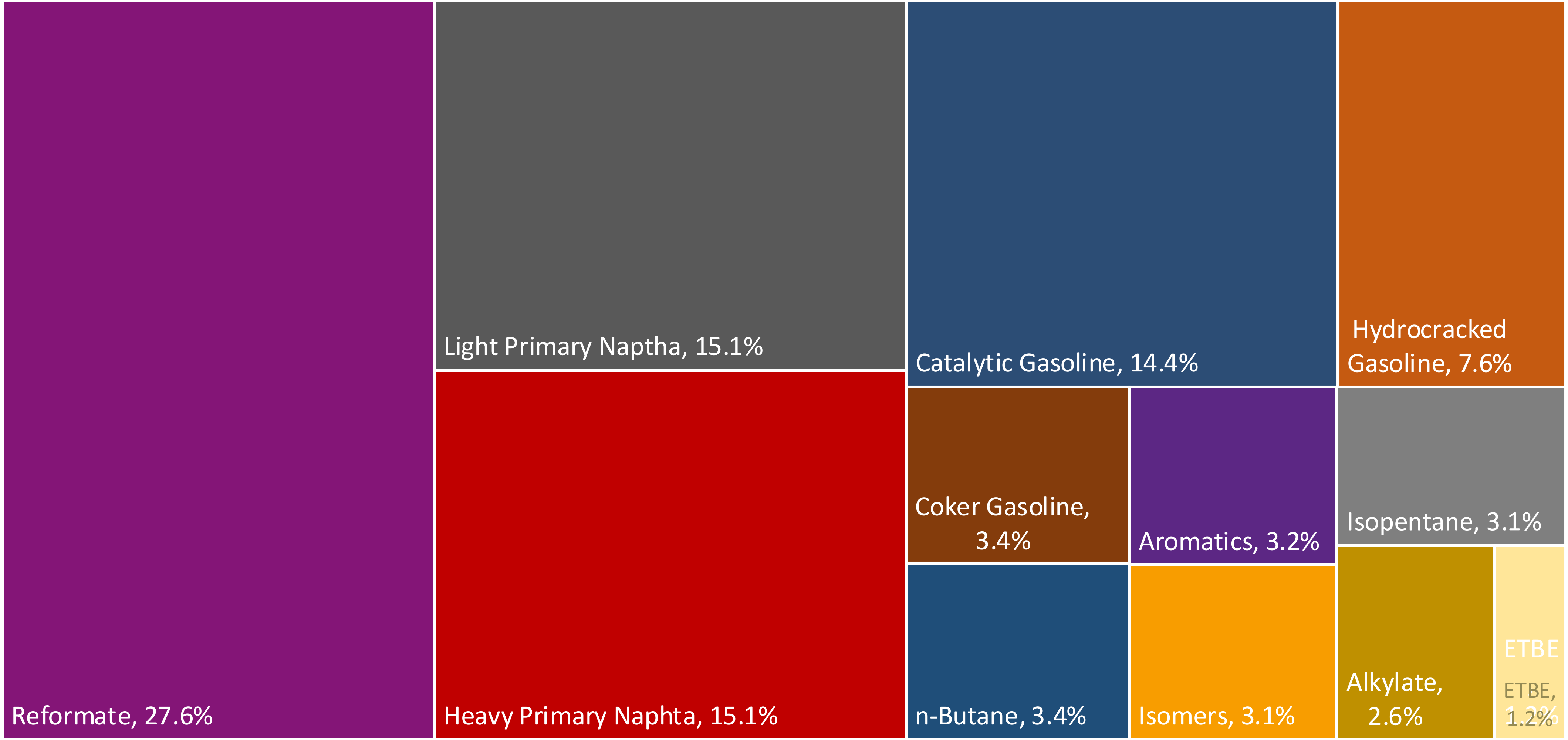
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

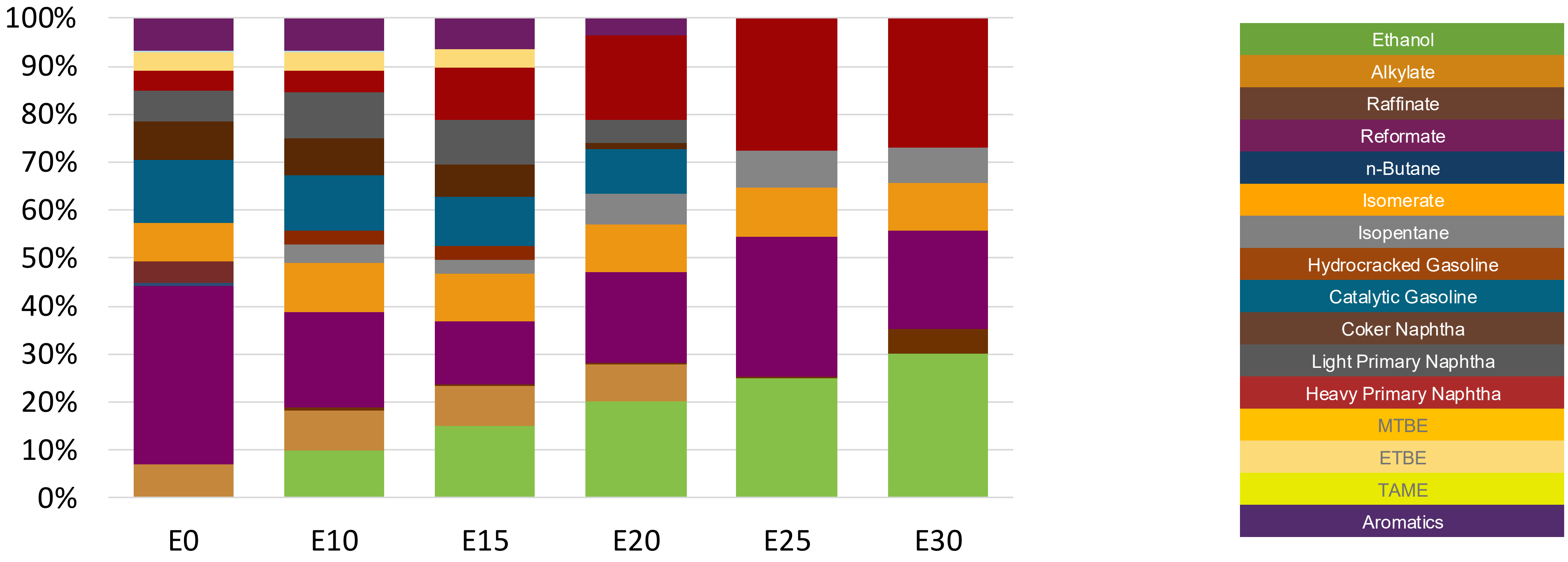


Spain Available Blending Components



Source: Faro90

Spain – Gasoline 95 – Constant Octane

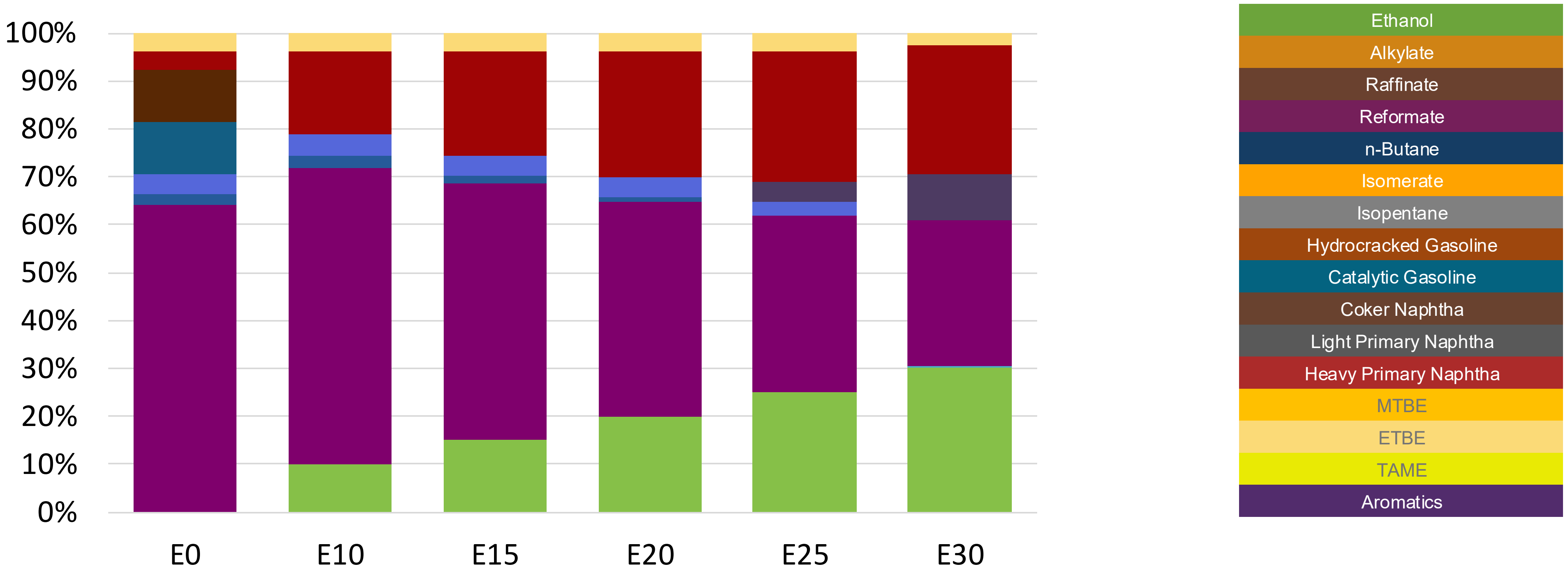


Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US$/gal)	$ 2.280	$ 2.168	$ 2.088	$ 1.998	$ 1.883	$ 1.820

Average CIF 2024 Prices

Source: Faro90

Spain - Gasoline 98 - Constant Octane

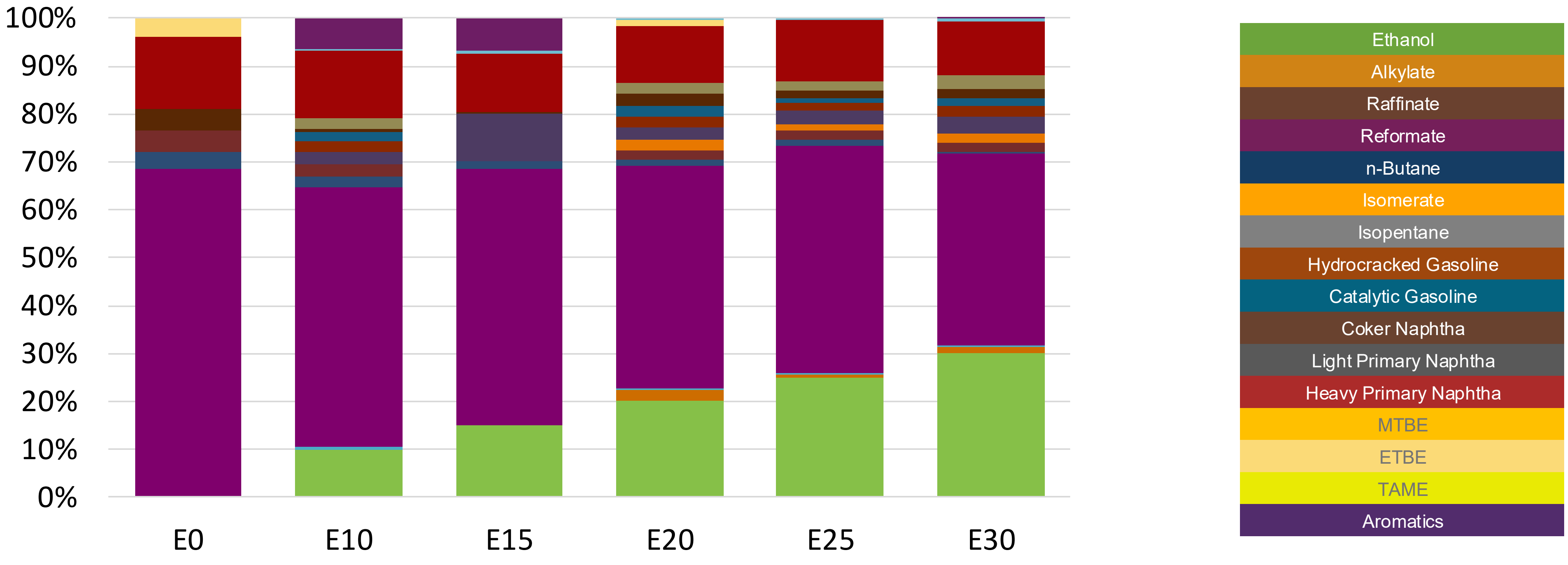


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.349	$ 2.225	$ 2.156	$ 2.081	$ 2.003	$ 1.924

Average CIF 2024 Prices

Source: Faro90

Spain – Gasoline 95 – Increased Octane

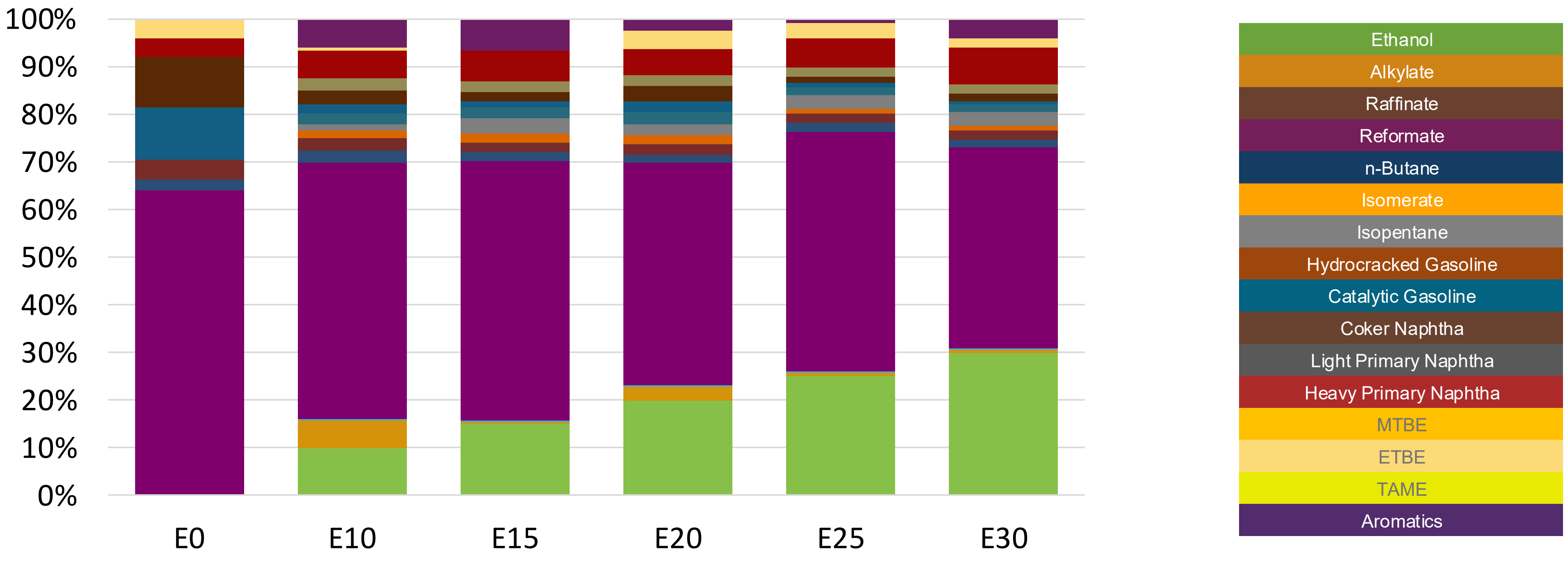


Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US$/gal)	$ 2.212	$ 2.212	$ 2.212	$ 2.212	$ 2.212	$ 2.212

Average CIF 2024 Prices

Source: Faro90

Spain – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US$/gal)	$ 2.349	$ 2.349	$ 2.349	$ 2.349	$ 2.349	$ 2.349

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

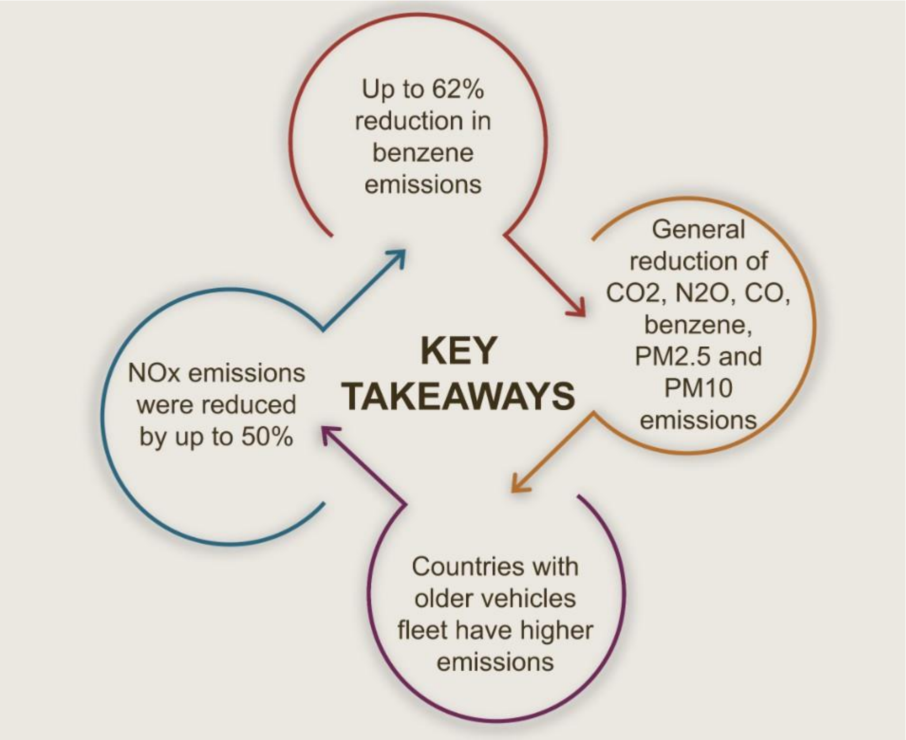
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - Spain

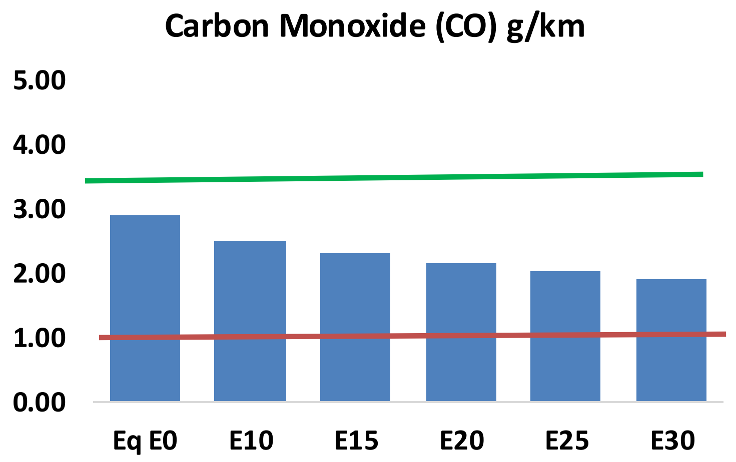
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	9,967
		Single FI	3-Ways		>8 years	184,079
		Multi FI	3-Ways		>8 years	1,986,509
			Euro III		>8 years	2,997,881
			Euro IV		>8 years	4,130,395
			Euro V		>8 years	1,048,814
			Euro 6		>8 years	2,531,411
					4-8 years	4,081,695
					<4 years	2,183,857
			Hybrid		All	771,450
		Electric			All	119,763
	Medium	Carburator	3-Ways	PCV Tank	>8 years	986
		Single FI	3-Ways		>8 years	5,454
		Multi FI	3-Ways		>8 years	291,511
			Euro III		>8 years	561,099
			Euro IV		>8 years	1,431,712
			Euro V		>8 years	321,320
			Euro 6		>8 years	843,804
					4-8 years	1,857,131
					<4 years	1,768,939
			Hybrid		All	538,150
		Electric			All	88,922
Sml Engines	Light	4-Cycle	None	PCV	>8 years	24,824
			Euro III		>8 years	80,432
			Euro IV		>8 years	603,608
			Euro V		>8 years	174,836
			Euro 6		>8 years	450,562
					4-8 years	623,207
					<4 years	754,983
			Hybrid		All	135,392
		Electric			All	21,575
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	-
		Fuel-Injection	3-Ways		>8 years	749,356
			Euro III		>8 years	951,622
			Euro IV		>8 years	1,363,585
			Euro V		>8 years	240,214
			Euro 6		>8 years	602,646
					4-8 years	792,932
					<4 years	632,043
			Hybrid		All	1,972
		Electric			All	141,946

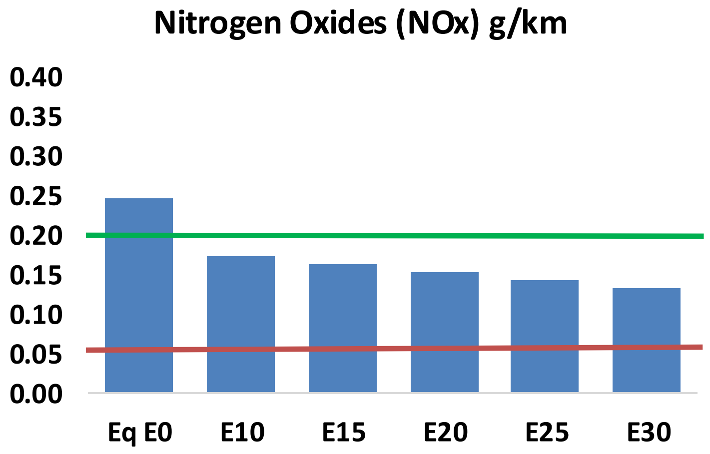
Vehicle Fleet: 36,100,582
Gasoline Vehicle Fleet: 11,736,365

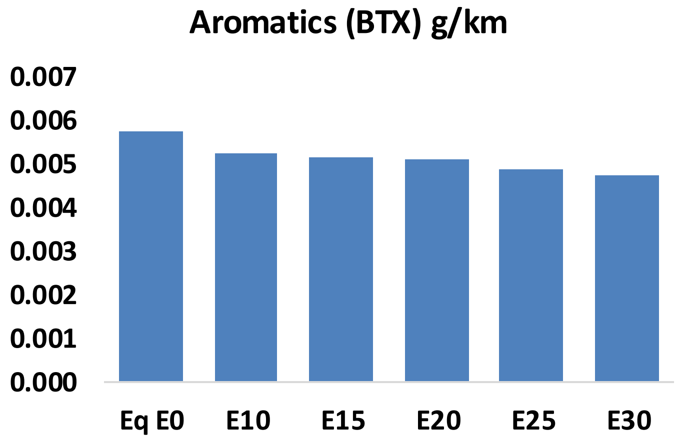
Source: Eurostat, ACEA

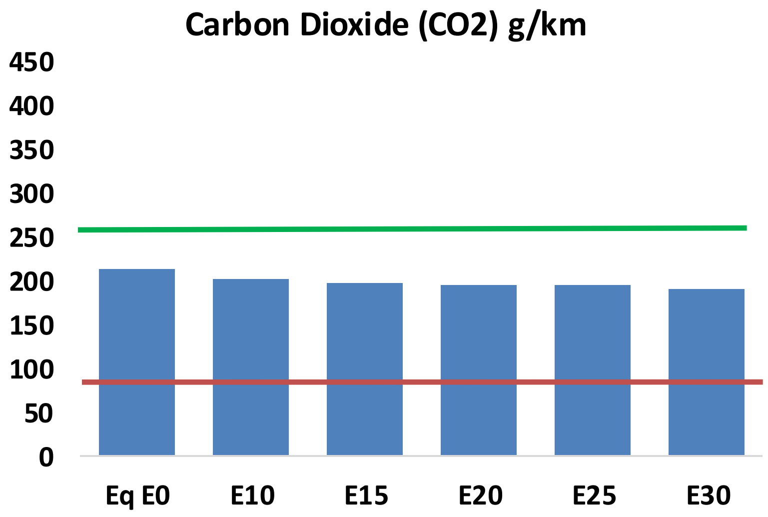
Spain – Vehicular Emissions

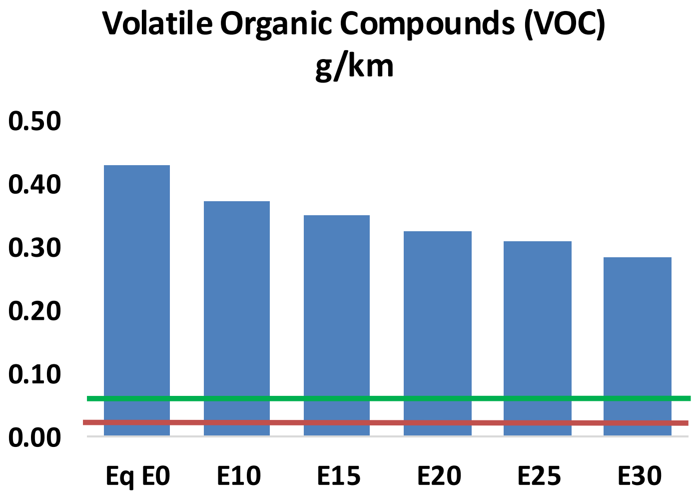
TIER III BIN 125 United States
Euro 6 Standard

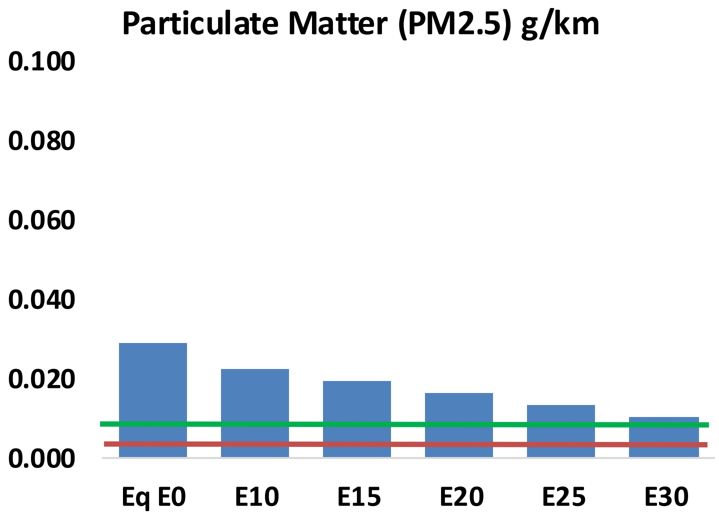












Source: Faro90

Spain – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	861,045	800,129	739,214	690,327	641,604	604,282	562,670	-14%	-25%
CO2	62,968,337	61,349,152	59,819,921	58,617,225	58,022,085	57,682,674	56,619,416	-5%	-8%
VOC	127,636	118,955	110,275	103,383	96,548	91,324	84,243	-14%	-24%
NOx	73,264	54,948	51,285	48,336	45,585	42,552	39,343	-30%	-38%
SOx	751	694	638	590	543	494	441	-15%	-28%
NH3	22,063	21,845	21,626	21,729	21,974	22,145	22,288	-2%	0%
N2O	1,015	1,010	1,006	1,011	1,031	1,043	1,055	-1%	2%
CH4	27,368	27,368	27,368	27,778	28,381	28,901	29,393	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	838	810	782	767	755	747	732	-7%	-10%
Acetaldehyde	2,231	2,561	3,757	5,722	7,796	8,908	10,526	68%	249%
Formaldehyde	7,772	7,556	8,808	10,342	10,835	11,987	13,049	13%	39%
Benzene	1,710	1,643	1,557	1,532	1,519	1,453	1,406	-9%	-11%
PM2.5	3,893	3,457	3,020	2,623	2,227	1,807	1,370	-22%	-43%
PM10	4,624	4,106	3,588	3,116	2,646	2,147	1,627	-22%	-43%

Source: Faro90

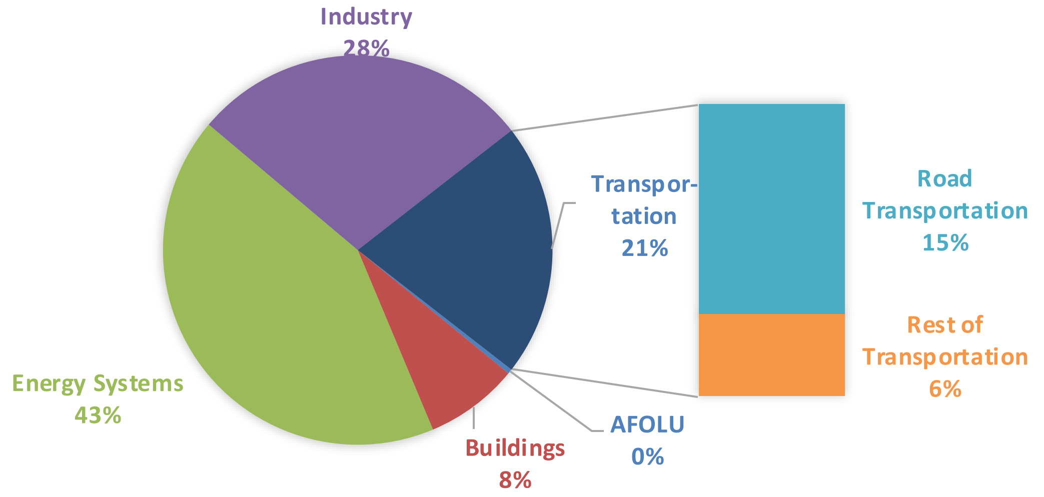
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

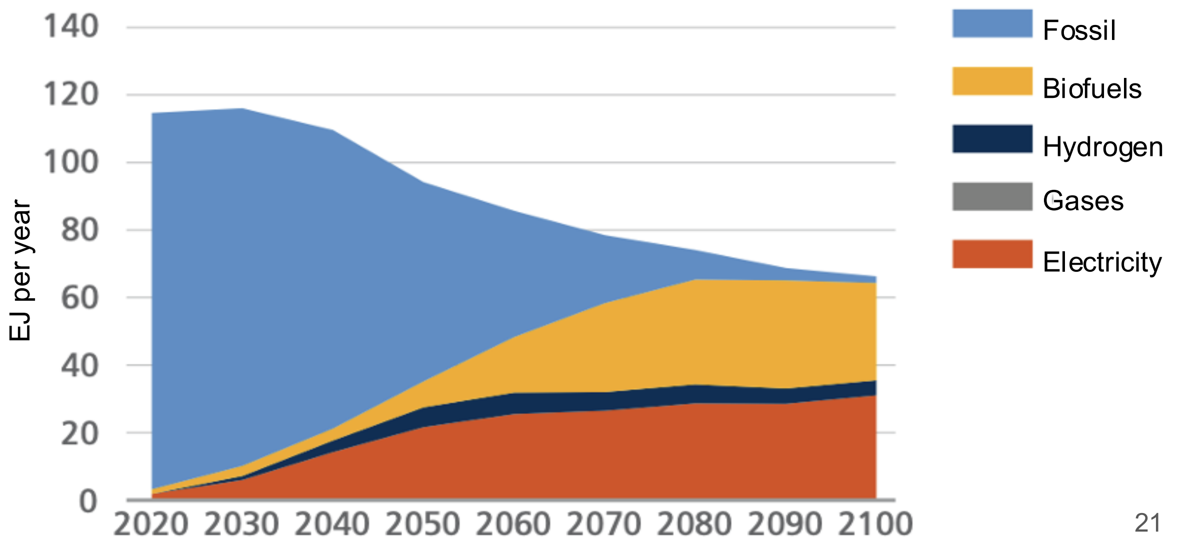
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

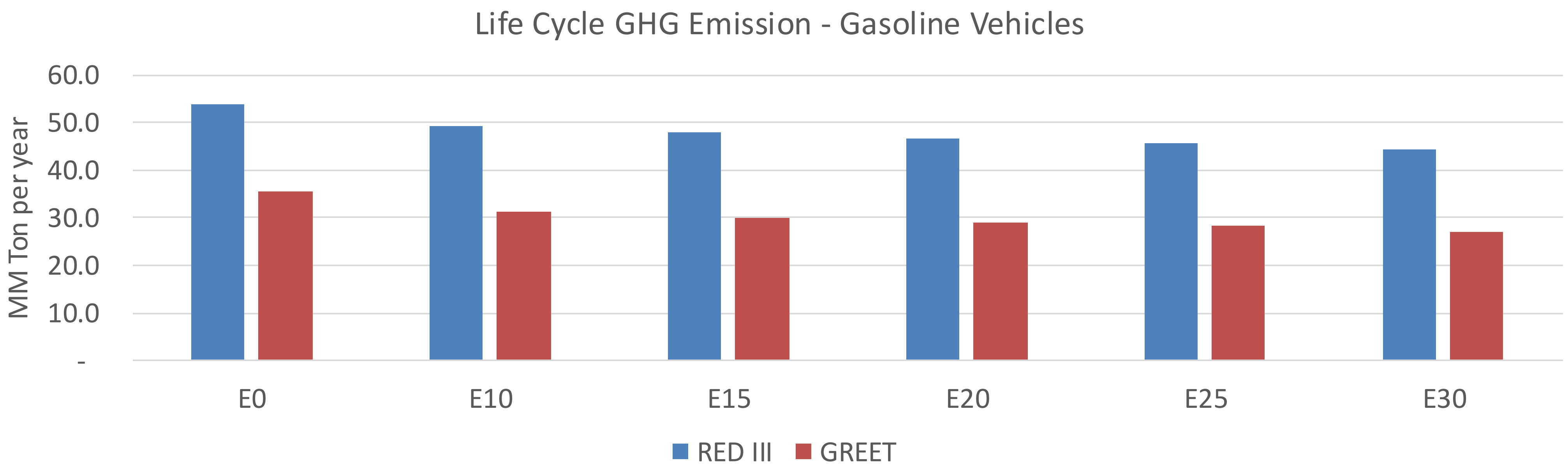
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Spain – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	269.9
Target @ 2035 (MMT/year)	151.9
Delta	118.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.1%	15.3%	17.9%	20.3%	23.6%
RED II/III	0.0%	8.8%	11.3%	13.3%	15.3%	17.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.6%	4.6%	5.4%	6.1%	7.1%
RED II/III	0.0%	4.0%	5.2%	6.1%	7.0%	8.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90